

CATEGORY		MID-RANGE INTEL MOTHERBOARDS				
	Brand	AOpen	AOpen	ASUS	Gigabyte	Gigabyte
	Model	MX45G-4DN	MX45G-N	P4P800-VM	GA 8IG1000MK	GA-8IPE1000-L
	Chipset	Intel 865	Intel 865	Intel 865	Intel 865G	Intel 865PE
	Interface type	Socket 478	Socket 478	Socket 478	Socket 478	Socket 478
FEATURES (25%)	Maximum CPU supported / FSB supported	P4 3.2 / 800	P4 3.2 / 800	P4 3.2 / 800	P4 3.2 / 800	P4 3.2 / 800
	Memory speed (MHz) / Max memory (GB) / Memory slot (nos)	DDR 400 / 4 GB / 4	DDR 400 / 4 GB / 2	DDR 400 / 4 GB / 4	DDR 400 / 4 / 4	DDR 400 / 4 / 2
	Form factor	Micro ATX	Micro ATX	Micro ATX	Micro ATX	ATX
	AGP slot / Onboard AGP / Type	✓ / ✓ / 8X	✓ / ✓ / 8X	✓ / ✓ / 8X	✓ / ✓ / 8X	✓ / ✗ / 8X
	PCI slots	3	3	3	3	5
	Onboard sound	✓	✓	✓	✓	✓
	Onboard FireWire / Firewire B	✗	✗	✗	✗	✗
	USB ports / Rear / Ports via cable	8 / 4 / 4	8 / 6 / 2	8 / 4 / 4	8 / 4 / 4	8 / 4 / 4
	Ethernet / Gigabit Ethernet	✓ / ✗	✓ / ✗	✓ / ✗	✓ / ✗	✓ / ✗
	SATA Connectors on motherboard / SATA RAID	2 / ✗	2 / ✗	2 / ✗	2 / ✗	2 / ✗
	Core voltage / Frequency adjustment	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓
	HyperThreading support	✓	✓	✓	✓	✓
	Component layout on the motherboard (S05)*	2.5	3	2.5	2	3
	Package contents					
	Manual / CDs	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓
	Necessary cables provided	✓	✓	✓	✓	✓
PERFORMANCE (75%)	Gaming Benchmarks : Serious Sam Second Encounter					
	Average fps	273.1	269.7	269.1	268.9	269.2
	UT2003 Flyby Demo					
	Average fps	337.24	335.75	339.15	335.77	336.86
	Quake III Arena					
	Average fps	476.4	534.7	525.1	505.6	502
	Real World Benchmarks					
	Zdbench Content Creation 2004	27.9	27.7	27.6	22.2	27.7
	Zdbench Business Winstone 2004	22.7	22.4	22.7	27.8	22.6
	Video Encoding (secs)	69	72	72	72	71
	Synthetic Benchmarks					
	3D Mark 2003 (patch 340)	3,399	3,393	3,396	3,392	3,390
	3D Mark 2001 SE	11,873	11,913	11,925	11,920	11,929
	PC Mark 2004					
	CPU Index	4,894	4,892	4,887	4,819	4,799
	Memory Index	4,605	4,468	4,474	4,487	4,497
	Graphics	2,791	2,789	2,792	2,842	2,783
	Disk Index	2,907	2,908	2,923	2,786	2,908
	Total PC Marks	4,905	4,907	4,907	4,898	4,950
	Sisoft Sandra 2004 Professional : CPU sub-system					
	CPU Dryhstone / Whetstone	9,791 / 4,013	9,472 / 4,003	9,736 / 3,796	9,773 / 4,013	9,812 / 4,023
	Multimedia - CPU Integer / FPU SSE	24,539 / 34,285	24,496 / 35,499	24,431 / 34,030	24,546 / 34,503	24,607 / 34,711
	Memory sub-system					
	ALU / RAM	4,634	4,362	4,380	4,379	4,386
	FPU / RAM	4,598	4,381	4,388	4,387	4,402
	Disk drive					
	Drive Index MB/sec	24,536	24,535	24,517	24,482	26,204
OVERALL SCORE	Performance	58.14	57.67	57.68	57.38	57.63
	Features	18.50	17.80	18.50	18.20	17.80
	Grand Total (Performance + Features)	76.64	75.47	76.18	75.58	75.43
	VFM (Index)	9.49	11.10	12.09	12.81	10.93
	Price (Rs)	8,075	6,800	6,300	5,900	6,900
	Grades	B+	B+	B+	B+	B+

*Scale of 5

ASUS PC-DL Deluxe (Dual Intel Xeon-based motherboard)

The PC-DL Deluxe board from ASUS is the only board based on the Intel 875P chipset for Intel's XEON range of processors. It's targeted at the workstation market, where Intel's Xeon processors are currently the norm.

The board supports two Xeon processors running at 533 MHz FSB. The memory controller can support up to 4 GB of DDR 333 MHz memory in dual-channel mode, along with unbuffered Error Correction Code (ECC). Theoretically, the dual-channel memory implementation provides the board with a peak bandwidth of 4.26 GBps for memory-intensive jobs.

Workstations are widely used in Computer-Aided Design (CAD) and Computer-Aided Manufacturing (CAM) applications. Such high graphic requirements are quenched by adding an 8X AGP Pro slot that can accept

any of the latest breed of graphics card.

The latest Southbridge ICH5R from Intel is at work on this board. So, you get native SATA support with RAID-0 and RAID-1 functions without the aid of a third-party controller. As far as I/O is concerned, ASUS has integrated a Promise SATA controller that offers RAID-0, 1 and 0+1 functions through two IDE ATA 133 and two SATA ports. This enables you to build a RAID array of two or four devices.

